

Data sheet



Customer item no.: P-101-3
Communication dated: 23/04/2024
Doc. no.: Quick quote
Quantity: 1

Number: ES 8002938088
Item no.: 100
Date: 23/04/2024
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AMAREX-IN F 50-220/032 ULG3-Ø140 S

Version no.: 1

Operating data

Requested flow rate	15.00 m³/h	Actual flow rate	15.14 m³/h
Requested developed head	19.00 m	Actual developed head	19.36 m
Pumped medium	Emulsion Water/oil Not containing chemical and mechanical substances which affect the materials	Efficiency	40.1 %
Ambient air temperature	35.0 °C	Power absorbed	1.98 kW
Fluid temperature	30.0 °C	Pump speed of rotation	2873 rpm
Fluid density	1000 kg/m³	Shutoff head	23.90 m
		Max. power on curve	2.90 kW
		Max. allow. mass flow	12.66 kg/s
Fluid viscosity	5.00 mm²/s	Design	Single system 1 x 100 %
Mass flow rate	4.21 kg/s		

Driver, accessories

Driver type	Electric motor	Motor enclosure	IP68
Drive standard mech.	KSB	Cos phi at 4/4 load	0.80
Model (make)	KSB	Motor efficiency at 4/4 load	76.0 %
Drive supplied by	Standard motor supplied by KSB - mounted by KSB	Temperature sensor	Bimetal switch 2x
Motor const. type	KSB Sub. motor	Motor winding	415V
Motor size	03	Number of poles	2
Motor speed	2873 rpm	Starting mode	Direct-on-line starting
Frequency	50 Hz	Connection mode	Star
Rated voltage	350-450 V	Motor cooling method	Surface cooling
Rated power P2	3.10 kW	Motor material	IS 210 GR. FG260
Available reserve	55.79 %	Frequency inverter operation allowed	FI allowed
Rated current	7.3 A	Cable length	10.00 m
Starting current ratio	6.85	Cable material no	IN623030
Insulation class	F to IEC 34-1		

Design

Pump standard	KSB-Aggregate international execution	Shaft seal type	FG
Design	Close-coupled submersible	Material code	SIC/SIC/NBR
Orientation	Vertical	Sealing plan	T Tandem mechanical seal
Suction nominal dia.	DN 50	Seal chamber design	Standard seal chamber
Suction nominal pressure	unmachined	Impeller diameter	140.0 mm
Suction flange drilled according to standard	EN1092-2	Free passage size	40 mm
Discharge nominal dia.	DN 50	Direction of rotation from drive	Clockwise
Discharge nominal pressure	PN 16	Bearing type	Anti-friction bearings
Discharge flange drilled according to standard	EN1092-2	Lubrication type	Grease
Shaft seal	Double-acting mechanical seal	Color	Ultramarine blue (RAL 5002) KSB-blue
Shaft seal manufacturer	KSB		

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Installation parts

Installation type	Stationary with guide wire	Type	Chain / Shackle
Scope of supply	Pump with installation parts	Material	Steel ST
Installation depth	4.50 m	Length	5 m / 1 PC
Material concept	G	Max. load	150 kg
Installation part kit	IN613218	Ident-number	IN623057
Ident-number			

Duckfoot bend

Size	DN 50	Guide wire	
Flange design	EN	Length	10.00 m
Duckfoot bend size (DN2 / DN3)	DN 50 Drilled according to EN	Material	A276 Type 304
Material	Grey cast iron EN-GJL-250	Ident-number	IN623048
Mounting type	Composite anchor bolts		
Foundation rail	Without		

Claw

Design	Straight
Size	DN 50

Lifting chain / -rope

Materials G3

Pump casing (101)	Grey cast iron EN-GJL-250/A48CL35B	O-Ring (412)	NBR 70 ACRYLNITR-BUTAD-KAUTS
Suction cover (162)	Grey cast iron EN-GJL-250/A48CL35B	Disc (550)	CH.70IRHD
Shaft (210)	Chrome steel 1.4021+QT800	Motor housing (811)	CrNiMo steel A4
Impeller (230)	Stainless steel A 743 Gr CF8M	Motor hous cover (812)	Grey cast iron EN-GJL-250/A48CL35B
Bearing housing (350)	Grey cast iron EN-GJL-250/A48CL35B	Cable bushing (834)	Grey cast iron EN-GJL-250/A48CL35B
Profile joint (410)	NBR	Hexagon socket head cap screw (914)	Brass CuSn10C Stainless steel A2-70

Nameplates

Nameplates language	International	Duplicate nameplate	With
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Performance curve

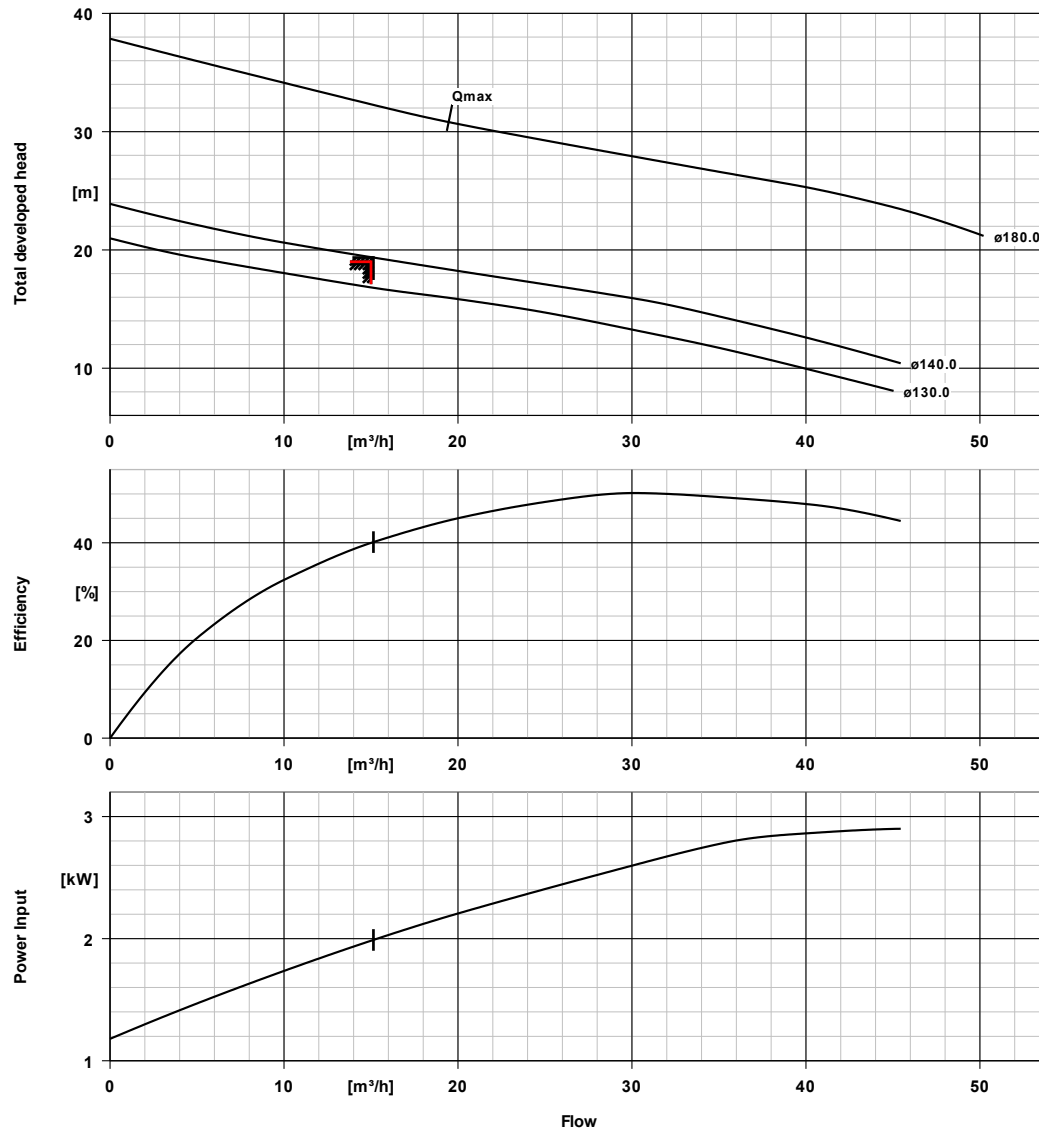


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Curve data

Speed of rotation	2873 rpm	Requested developed head	19.00 m
Fluid density	1000 kg/m³	Efficiency	40.1 %
Viscosity	5.00 mm²/s	Power absorbed	1.99 kW
Flow rate	15.14 m³/h	Curve number	K2563-52-04S
Requested flow rate	15.00 m³/h	Effective impeller diameter	140.0 mm
Total developed head	19.36 m		

Motor data sheet



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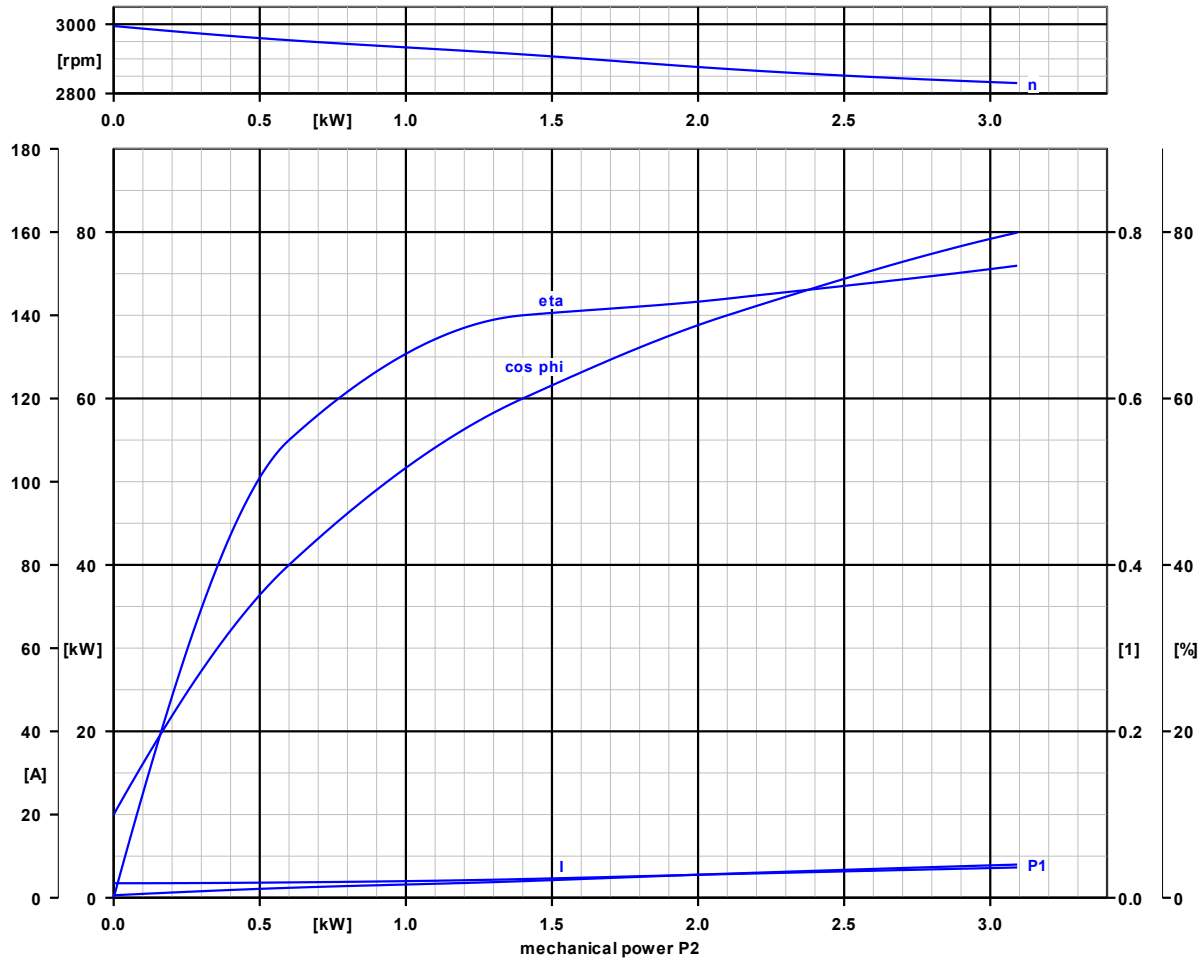
Motor data

Motor manufacturer	KSB	Rated speed	2830 rpm
Motor size	03	Starting current ratio	6.84931507
Motor construction type	KSB Sub. motor	Starting mode	Direct-on-line starting
Motor material	IS 210 GR. FG260	Power cable	H07RN-F 7G1.5
Efficiency class	not classified	Number of power cables	1
Rated voltage	350 - 450 V	Power cable Ø min.	14.7 mm
Frequency	50 Hz	Power cable Ø max.	16.7 mm
Motor power	3.10 kW	Cable standard	Without
Rated current	7.3 A		

Curve data

The no-load point is not a guarantee point within the meaning of IEC 60034

Load	0.0 %	19.4 %	45.2 %	67.7 %	100.0 %
P2	0.00 kW	0.60 kW	1.40 kW	2.10 kW	3.10 kW
n	2995 rpm	2954 rpm	2913 rpm	2871 rpm	2830 rpm
P1	0.30 kW	1.20 kW	2.00 kW	2.90 kW	4.00 kW
I	3.5 A	3.7 A	4.5 A	5.7 A	7.3 A
Eta	0.0 %	55.0 %	70.0 %	72.0 %	76.0 %
cos phi	0.10	0.40	0.60	0.70	0.80



Installation plan

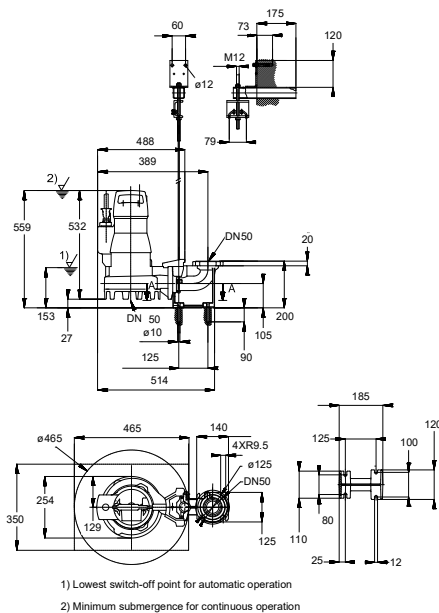


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Drawing is not to scale

Dimensions in mm

Motor

Motor manufacturer	KSB
Motor size	03
Motor power	3.10 kW
Number of poles	2
Speed of rotation at duty point	2873 rpm

Connections

Suction nominal size DN1	DN 50 / EN1092-2
Discharge nominal size DN2	DN 50 / EN1092-2
Nominal size DN3	DN 50 / EN
Nominal pressure suct.	unmachined
Rated pressure disch.	PN 16

Weight net

Pumpset (Motor + Cable)	68 kg
Installation parts kit	8 kg
Total	76 kg

Connect pipes without stress or strain!